

nile.

COFFEE INTO CLIMATE ACTION

Give coffee a next life.

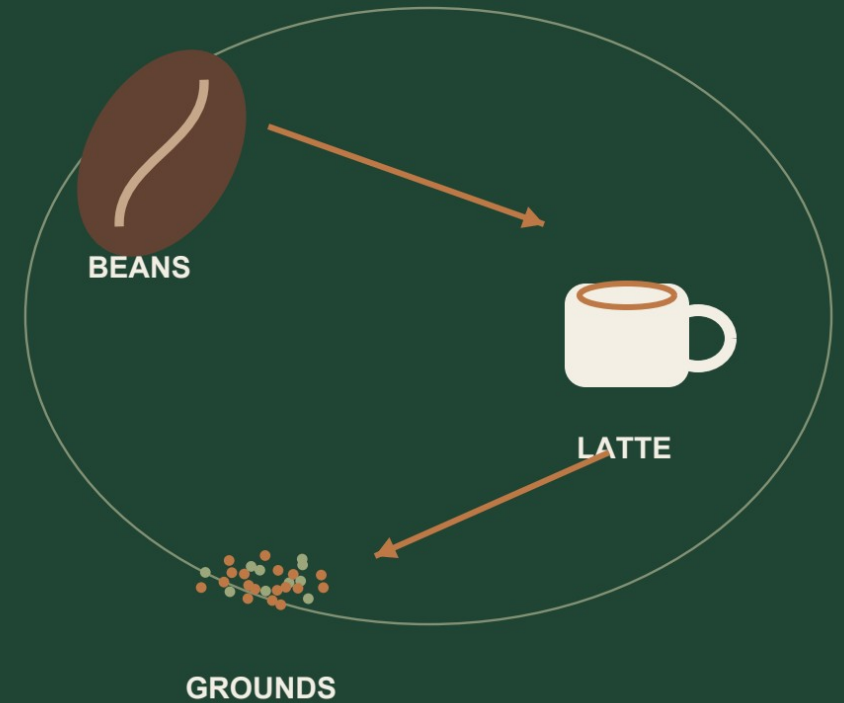
Surplus beans and coffee by-products.
Compatible recipients. Clear value. Easier collection.

~75,000 tonnes / year

Spent coffee grounds generated in Australia.
RMIT estimate, 2023. Context, not Nile impact.

Zero Waste & Methane Reduction / Green Industrialisation / Awareness

Source: RMIT, Coffee offers performance boost for concrete, 23 August 2023



One engine for coffee materials
at different stages.

The barrier is reliable reuse, every week.

A cafe needs an accepting destination, workable costs and a dependable handover.

4

cafes interviewed

Team-reported
qualitative feedback

Recurring spent grounds

Grounds commonly go in bins; actual disposal destinations need checking.

Occasional surplus beans

Usable leftovers sometimes need another buyer or cafe before freshness declines.

Inconsistent arrangements

Some cafes already use agricultural collection; others lack a repeatable system.

Early problem evidence only: no measured volumes, verified landfill routes or commitments to pay.

Three priorities. One practical workflow.

For Australian cafes, accepting processors and collectors: make coffee circularity easier to adopt and measure.

01

Zero Waste & Methane Reduction

Keep usable beans circulating; redirect eligible grounds from landfill.

Test: additional accepted mass and the actual disposal baseline.

02

Green Industrialisation

Help suitable growers and processors use coffee by-products as inputs.

Test: processor acceptance, actual use and input substitution.

03

Awareness

Explain reuse routes, preparation rules and climate estimates in plain language.

Test: understanding and correct preparation before and after use.

List. Match. Collect. Confirm.

One engine demonstrates spent grounds and usable surplus roasted beans.

01 Describe the batch

Type, condition, quantity, packaging and availability.

30 kg

proposed

02 Compare eligible matches

Check acceptance rules first; compare net value, distance and timing.

27 kg

accepted

03 Agree collection

Shared preparation and pickup brief; recipient accepts the current version.

25 kg

reported used

04 Record what happened

Actual accepted weight, supplier confirmation and separate reported use.

Demo example • separate evidence states

Less uncertainty. Clearer value.

The adoption hypothesis: reduce coordination effort without hiding handling or collection costs.

For the cafe

Repeat listings, understandable net value and a clear preparation checklist.

For recipients / collectors

Agree material, container, access, time and collection responsibility before pickup.

A reason to return

Usage / sales CSVs help plan bean purchases against stock, goals and budget.

Net benefit = sale revenue + documented avoided costs
– extra handling, collection and platform costs.

Reground proves collection is valuable.

Nile proposes batch-level choice, usable-bean matching and coordination; the commercial advantage needs testing.

Dimension	Reground: documented service	Nile: local prototype
Material routes	Grounds and chaff collection; wider circular projects.	Grounds + usable beans to sample cafes; wider catalogue unvalidated.
Collection	Bins and physical pickups.	Compatibility, cost comparison and shared collection brief; no fleet.
Evidence	Pickup/destination tracking, online reporting and updates.	Accepted weights, two-party receipts and separate reported use.
Business adoption	Education, consultancy and Circular Coffee ESG/certification.	Repeat listings, batch economics and CSV purchase planning.

Potential collector partner only; no agreement or proven price/climate advantage.

Sources: reground.com.au/collection/ground-coffee/ • [/collection/chaff/](#) • [/collection/circular-coffee/](#) • [/consultancy/](#)

What could 100 cafes contribute?

Assumed one-year cohort; only materials that would otherwise go to landfill are eligible.

83.2

tonnes of grounds

diverted per year

4.16

tonnes of usable beans

redirected per year

~6.2

tonnes of methane

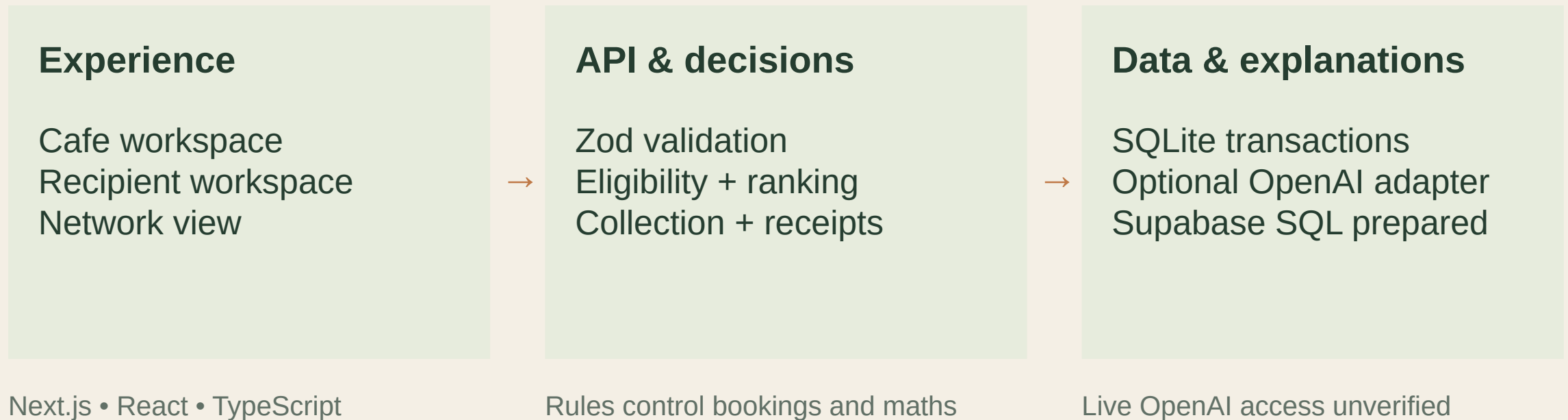
lifetime avoidance from
that year's grounds

Assumptions: 20 kg grounds + 1 kg beans/cafe/week • 52 weeks • 80% additional diversion • compost route • no landfill gas capture.

Generic food-waste methane proxy; no bean methane credit. Source: DCCEEW, NGA Factors 2026, Tables 15, 44 and 38.

Working locally. Cloud setup comes next.

GCP is not configured. Intended route: Next.js on Vercel + Supabase; neither is deployed.



Documented local validation: 48 unit/database + 13 browser/API checks. Live AI, cloud Auth and real recipients remain unverified.

A small pilot can test the whole promise.

Proposed: 3–5 nearby cafes • 1–2 accepting recipients • 4 weeks. No participants are confirmed.

1 Establish the baseline

Check bin destinations, existing collectors, weekly weights and current costs.

2 Agree a practical route

Confirm recipient rules, storage, pickup windows, access and charges.

3 Run and record

Track accepted kg, reported use, failures, staff time, costs and understanding.

Suggested pass gates

≥80% pickups completed • ≤2 min median repeat listing • ≥60% repeat participation
Median net benefit ≥\$0 after actual costs; additional diversion checked against baseline.

Assumptions stay visible and adjustable.

The four interviews did not measure these quantities. The 100-cafe case is a planning illustration.

Input / calculation	Assumed value / result
Cohort and period	100 cafes × 52 weeks × 80% additional successful diversion
Eligible grounds and usable beans	20 kg grounds + 1 kg beans per cafe per week
Annual material	83.2 t grounds + 4.16 t beans = 87.36 t
Grounds-only methane formula	$83.2 \times (2.1 \times (1 - \text{capture}) - 0.021) \div 28$
Assumed landfill methane capture	0%: 6.2 t CH ₄ 50%: 3.1 t 70%: 1.8 t

Methane is a lifetime comparison, not a measured annual saving. Generic food-waste proxy; beans receive no methane credit. Transport and processing energy are excluded.

Source: DCCEEW 2026, Australian National Greenhouse Accounts Factors, Tables 15, 44 and 38 (CC BY 4.0).

What we used, and what is still planned.

Full disclosure, setup instructions and source links are included in the repository.

AI assistance	OpenAI Codex: planning, research, writing, coding, review and documentation.
Application	Next.js, React, TypeScript, Tailwind, Zod, Radix/CVA, Recharts, Lucide and SQLite.
Validation & coordination	Vitest, Playwright, PGlite, ESLint, Prettier, agent-browser, GitHub and Notion.
Submission materials	Artifact Tool, LibreOffice, pypdf and pypdfium2; earlier PDF used ReportLab.
Prepared, not live	OpenAI adapter/key locally configured; live access unverified. Supabase SQL only; no cloud deployment.